

Navigators Technical Documentation

Navigators — Dataset & Data Provenance Specification

Document Identifier: DATA-NAV-01
Status: Production Reference
Version: 1.0.0
Last Updated: 2026-09-23
Attribution: Developed by Navigators

1. Data Classification Framework

To maintain scientific integrity and respect data licensing, Navigators establishes a transparent, three-tier classification of all datasets utilized in the repository:

1. PUBLIC EXTERNAL BENCHMARK DATASET
• IO-VNBD (Inertial and Odometry Vehicle Navigation Benchmark) • Authors: Onyekpeu et al. (Open Access / Research Use) • Provenance: External public repository (Git LFS)
2. DERIVED & SYNTHETIC DATASETS
• Kinematic simulated vehicle trajectories • Preprocessed and windowed feature tensors (105,706 windows) • Provenance: Formatted via `src/data/` data preparation pipeline
3. NAVIGATORS INGESTED TELEMETRY (OWN COLLECTED)
• Crowdsourced smartphone drives recorded via browser client • Provenance: User-consented local recordings via `data_recorder.js` • Current Status: Prototype diagnostics (`data/raw_trips/`)

2. External Dataset: IO-VNBD Benchmark

2.1 Attribution & Citation

- **Full Title:** Inertial and Odometry Vehicle Navigation Benchmark Dataset (IO-VNBD)
- **Creators / Authors:** Uche Onyekpeu et al.
- **Repository:** <https://github.com/onyekpeu/IO-VNBD>
- **Ownership Notice:** Navigators claims zero proprietary ownership over IO-VNBD. It is utilized strictly under open scientific benchmarking terms.

2.2 Dataset Structure & Sensor Payloads

The dataset consists of synchronized CSV pairs recorded across multiple driving sessions: - `S_*.csv` (**Sensor Stream**): High-rate vehicle sensor readings. - Accelerometer: a_x, a_y, a_z in meters per second squared

(m/s²). - Gyroscope: $\omega_x, \omega_y, \omega_z$ in radians per second (rad/s). - Magnetometer: μ_x, μ_y, μ_z in microteslas (μT). - **V_*.csv (Ground Truth Kinematics)**: High-precision GNSS/INS reference solution. - Linear Speed: Velocity magnitude in meters per second (m/s). - Ground Track: True heading in degrees ($^{\circ}$). - Cartesian Velocities: Converted to North (v_N) and East (v_E) coordinates:

$$v_N = \text{Speed} \times \cos(\text{Heading}_{rad})$$

$$v_E = \text{Speed} \times \sin(\text{Heading}_{rad})$$

2.3 Sampling Frequency & Resampling

Raw IO-VNBD sensor files sampled at 100 Hz are downsampled to a uniform 10 Hz rate using an anti-aliasing low-pass filter to match the nominal polling frequency of smartphone browser runtimes.

3. Session-Level Partitioning & Leakage Elimination

To ensure strict zero-leakage validation, the 144 independent driving sessions in IO-VNBD are partitioned at the file/session level:

Split	Number of Sessions	Total Windows ($W = 200$)	Fraction	Purpose
Train	100 sessions	63,423 windows	70.0%	Model parameter optimization.
Validation	22 sessions	30,349 windows	15.0%	Early stopping and hyperparameter tuning.
Test (Held-Out)	22 sessions	11,934 windows	15.0%	Final generalization benchmark evaluation.
Total	144 sessions	105,706 windows	100.0%	Comprehensive vehicle dynamics.

4. Derived Synthetic & Diagnostic Datasets

Located in `data/synthetic_trajectories/` and `data/real_dataset/`: - **Synthetic Vehicle Trajectories**: Mathematically simulated vehicle paths (constant acceleration, circular turns, stop-and-go maneuvers) corrupted with Gaussian noise and simulated accelerometer/gyroscope bias drift. Used in automated unit testing (`tests/test_ekf.py` , `tests/test_dead_reckoning.py`) to verify estimator convergence under known ground truth. - **Trip 4 Diagnostic Log (`data/raw_trips/trip_4.json`)**: Early phone recording used for file format validation and recorder pipeline testing. Note: As documented in `docs/replay-and-evaluation.md` , Trip 4 exhibits non-monotonic GPS timestamps and is restricted to diagnostic format checks, not scientific accuracy claims.

5. Ingested Smartphone Contributor Data Protocol

The mobile web client incorporates an integrated sensor logger (`simulator/data_recorder.js`): 1. **User Consent:** Prior to recording, users must explicitly grant location and motion permissions and toggle the dataset consent switch (`consent: true`). 2. **Recorded Fields:** - `timestamps` : Microsecond-resolution elapsed time. - `accel` : Tri-axial acceleration including Earth gravity (m/s^2). - `gyro` : Tri-axial rotation rate (rad/s). - `gnss` : Array of *[lat, lon, altitude, speed, heading, accuracy]*. - `metadata` : User-agent string, device platform, browser version, and SHA-256 integrity hash. 3. **Data Quality Verification:** The backend `DatasetRepository` executes an automated validation pass before accepting sessions into the training queue: - Rejects sessions with time reversals ($t_k \leq t_{k-1}$). - Rejects sessions with sensor dropout gaps > 1.0 s. - Rejects unphysically large accelerations ($> 50 \text{ m/s}^2$).

Developed by Navigators